# ML22 The final run: your project, ready to share

Week 8, Hand over | Session 22 of 24 | Machine Learning | New library: sklearn

Today's question. How well does my frozen model do on rows nobody has used?

Objective. By the end: you can run the held-back check once, report it separately from validation, and share a project that reruns from a restarted kernel.

> **WHERE THIS IS USED**
> Before a model goes live, a team freezes it, tests it once on untouched data and packs the work so someone else can rerun it. Today you do the same. Could a stranger rerun your project tomorrow?

Before this session: ML21 Compare, explain and critique. If you cannot explain the last example, use its error card with the instructor first.

## Words for today

| Word | What it means | Think of it like this |
|---|---|---|
| held-back run | The one honest score, run once after all choices are fixed. | Opening the sealed envelope. |
| frozen model | A model whose inputs and settings are fixed before the held-back run. | A recipe you stop changing before you serve it. |
| handover | Everything another person needs: data origin, code, instructions and limits. | Passing on the keys with directions. |
| restart and run all | Restart the kernel and run every cell from the top. | Starting the game from the first level. |
| limit | A fair statement of what the result cannot show. | The edge of what you know. |

## Block A. Run the held-back check once

The fares model with three inputs was chosen on validation. Now run the 30 held-back trips once.

Predict. Predict whether the held-back error will be above or below the validation error of 2255. Write your guess on paper first.

*Your guess, written before you run it*

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.dummy import DummyRegressor
from sklearn.metrics import mean_absolute_error
fares = pd.read_csv("synthetic_fares.csv")
X = fares[["distance_km", "rush_hour", "raining"]]
y = fares["fare_ugx"]
```

```
X_dev, X_test, y_dev, y_test = train_test_split(X, y, test_size=0.2, random_state=7)
X_train, X_val, y_train, y_val = train_test_split(X_dev, y_dev, test_size=0.25,
    random_state=7)
from sklearn.linear_model import LinearRegression
baseline = DummyRegressor(strategy="mean").fit(X_train, y_train)
model = LinearRegression().fit(X_train, y_train)
final_mae = mean_absolute_error(y_test, model.predict(X_test))
final_baseline_mae = mean_absolute_error(y_test, baseline.predict(X_test))
print(round(final_mae), round(final_baseline_mae))
```

Run it, then compare. The cell prints 2936 and 3617. On the 30 held-back trips the model misses by about UGX 2936 and the plain guess by about UGX 3617. The model is still better, but its lead is much smaller than the validation lead of 2255 against 4361. This is normal: the held-back set has only 30 trips, and the validation lead may have been lucky. Report the final number as it is, next to the validation number, and do not change the model to improve it.

Investigate.

1. Why report the final score separately from validation?
2. Can you change the model if the final score is disappointing?

Modify. Calculate how much smaller the model's error is than the baseline's on the held-back trips, as a percentage.

## Block B. The handover checklist

Another person should be able to rerun your folder. Work through the list with a partner.

Predict. Write the three things you think a stranger needs first to rerun your project. Write your guess on paper first.

*Your guess, written before you run it*

Run it, then compare. The checklist: restart the kernel and run all cells from the top; the data file sits beside the notebook; the data origin and the fact that it is synthetic are stated; the held-back result is reported separately from validation; limits are written next to the result; project.ipynb, the data dictionary and report.pdf or report.docx are saved together; every file opens when reopened. A model on invented teaching data is a demonstration, and nobody should use it to make decisions about people.

Investigate.

1. Why restart the kernel before the final run?
2. What does the report say about the model's limits?

Modify. Swap notebooks with a partner. Follow their instructions without asking questions and note where you stall.

## Make

Use the project checklist. Reopen all your files. Save project.ipynb, your data dictionary and report.pdf or report.docx together in your project folder, and share the folder with the instructor by the route in START_HERE.pdf.

Save your work as ML22. Keep the worked example and put your own attempt below it. Write two sentences: what you changed and how you checked it.

## Error card

| You see | It means | Fix |
|---|---|---|
| NameError after restarting the kernel | A cell used a name that was only created in a deleted or skipped cell. | Run all cells from the top and add the missing step. |
| The final score differs from last time | The split or the model settings changed. | Use the same random_state and settings as in validation. |

## Exit ticket

The held-back set is used once, reported next to validation, and the project reruns from a restarted kernel.

AI use follows AI_USE_POLICY.pdf. In weeks 1 to 6 no AI-generated solutions are used for exercises, labs or build tasks. Practise for 20 to 30 minutes between classes by repeating the Make task with one value changed. If you are stuck for 30 minutes, save the error and bring it to class. A file saved on your laptop has not been shared until you use the route in START_HERE.pdf.